



As climate variability continues to impact Australian agriculture, farmers are looking for proven ways to mitigate the impacts by building resilience into their businesses.

Under the Future Drought Fund's Long-term Trials program, different scenarios for crop and livestock production in mixed farming systems are being explored to determine how they can improve drought preparedness. The project is a collaborative, long-term research project, led by the University of Melbourne in partnership with the University of Tasmania, Federation University and grower groups across Victoria.

The project uses a hub and spoke model with highly monitored, replicated trials at the University of Melbourne's Dookie Campus, complemented by satellite trials and multiple on-farm demonstration sites. This approach ensures the research is scientifically rigorous while remaining practical and regionally relevant. Farmers have been involved from the outset, helping shape the research questions to reflect real-world needs and reduce risk when adopting new practices.

Long-term Trials Project Co-Lead Professor James Hunt explains, "Most broadacre businesses in Australia use a combination of crops and livestock, and what we want to understand is how changing the ratio of those two enterprises affects drought resilience." The findings from these trials will support farmers to make evidence-based decisions about enterprise mix and technology adoption, improving efficiency and profitability while safeguarding natural resources.

"Between now and 2050, we've still got the most rapid period of population growth that there ever has been on the planet, and we need to produce enough food for those people," says Professor Hunt.

“Growers need sustainable farm businesses that deliver profit without damaging the environment, and research like this helps identify practices that meet those goals.”

By reducing uncertainty and providing regionally specific data over several growing seasons, the trials aim to give farmers more confidence in adopting practices to prepare for tough seasons, build drought resilience into their mixed farming enterprises and adapt to changing conditions.

Find out more information and get project updates about this long-term trial, via the [Victoria Drought Resilience Adoption and Innovation Hub website](#).

Watch the video to see how the University of Melbourne and their project partners are leading research trials to build drought resilience in farming systems.

Video duration: 9 mins 32 secs

Introduction

This is the transcript of a video case study produced by the Australian Government's Future Drought Fund (FDF). The University of Melbourne received funding, under the FDF's Long-term Trials of Drought Resilient Farming Practices Grants Program, for their project - Long-term economic, environmental and social outcomes of drought resilience practices in mixed farming. The video shows how the University of Melbourne, in collaboration with their research and farming systems group partners, are seeking to provide rigorous scientific evidence to improve the drought resilience of broadacre grains, grazing and mixed farming systems.

Learn more about the [Future Drought Fund](#)

[Recording begins]

Music plays.

Professor James Hunt [00:32]

Our long-term trial project is about comparing the intensity of crop production with livestock production in broadacre farming.

Most broadacre businesses in Australia use a combination of crops and livestock, and what we want to understand is how changing the ratio of those two enterprises affects drought resilience.

The project is collaborative between the University of Melbourne, University of Tasmania and Federation University, and we also have grower groups in the Birchip Cropping Group in western Victoria, Southern Farming Systems in southern Victoria, and Riverine Plains here in northeast Victoria, and there's two reasons why it's important to collaborate. The first is that the expertise you need to solve a complex problem are rarely in one organisation, and so you need to source the expertise from multiple organisations to achieve what you want to achieve.

The second reason is that we want to reach a diverse range of environments with our extension and our message, and having locally relevant trials and locally relevant groups that can engage with farmers and advisors is important for the success of the project.

The whole project is set up with a hub and spoke model where we have central, highly monitored, scientifically rigorous experiments, like the one we've got here at Dookie College. We've got three different treatments and three replicates of those treatments. But in order to get the findings of those out into the wider area, we have demonstration sites with grower groups where aspects of what we're implementing in the hub model are demonstrated in cooperating growers' paddocks.

We're very anxious to try and make sure that this research is relevant and usable, and from the outset we've tried to design the research in collaboration with farmers in a participatory kind of way, so all the experiments were established with consultative groups where we got local farmers and advisors to sit down with us and pose the questions that they really want answered in terms of their mixed farming business structure, and that was different in different regions, and that's good because if we want things to be regionally relevant, we need to adjust our experiments so that they're different in the different regions and address the questions in those parts of the world so that it's easier to adopt and work into a business structure, and that it gives growers the confidence to adopt.

Today, we had our annual field day at Dookie Campus where we want to showcase all the research that we have, but particularly focus on our long-term trial because we consider that a bit of a jewel in the crown of our field research that we have at Dookie Campus, and we invite all and sundry, but the people that we really want to attract are local growers and advisors to get that investment and buy-in into what we're researching here, and also help guide our research to make sure that it's relevant and adoptable.

Robert Brown [03:36]

It's important for farmers to be involved in the co-design of the long-term trials so that we get accurate information on what we're trying to change already. Most farmers, aren't doing anything that they did 25-30 years ago. We've all changed our practices, we've all changed our varieties and we just need quantified information done at a local level so we can make sure we're making the right changes and we're heading in the right direction.

Matthew De Roos [04:01]

The benefits of running on-farm scale trials with growers is that they're able to see the results in real time in their paddocks, and it's also for their neighbours and locals as well, who can see the results come through. By the end of it, they know that it's done in a large area and we're not trying to do it in a small plot, and it just provides validity to what we're doing and shows that the progression through the trial as well.

Rhiannan McPhee [04:27]

Understanding what different rotations could mean for preparing for drought and various species of crops and how to integrate livestock better and putting that on a long-term scale is something that we don't often get to see in research trials, so this project has, you know, enabled that for our regions and being able to replicate across the state it means we can share learnings from, you know, if one part of the state might be dry, where the other part might not be.

You can, you know, understand how that would work in those different regions. You can also collaborate with other projects that are doing a similar thing on this, which is the benefit of being involved with Future Drought Fund projects.

Professor James Hunt [05:04]

Evidence-based trials are important because growers' terms of trade are declining all the time. So, in order to stay ahead, they need to increase their efficiency of production and lower their cost of production, and the best way that you can do that is by adopting new technology.

So the best and most efficient way that you can adopt what's gonna work on your farm and your farming business is by looking at the evidence, and the best form of evidence is from replicated field experiments that have been done in your region.

Being a drought trial, we're very interested in measuring water balance and we're lucky to have three Eddy Covariance Flux Towers in the trial so we can get a very accurate estimate of the water balance of each of our systems in real time. One of the other tools that we're using is capacitance moisture probes that lets us continually measure the soil water status under the different crops and in the different systems so we know how the previous season's crop and performance of that crop or pasture relates to the availability of water in the next season, and that's a important thing when you're managing drought, is carrying water from one season into the next.

We also use hyperspectral imaging in fixed-wing aircraft to estimate the growth in our different treatments, and that lets us build up a map of how this is spatially varying in our field experiment.

Matthew De Roos [06:35]

Building drought and climate resilience within farming systems is there to protect them against the changing landscapes and just safeguard for future as well.

My role within this trial is that I'll come out and organise any sampling to be done, soil samples, field mapping as well, where we're doing grazing, the hay, I'll do plant cuts, biomass cuts, and just provide all those results back into our system, and then back to the reports for the funding partners.

Professor James Hunt [07:04]

The secret to success of a good long-term trial project is to set up your treatments in such a way that the systems can continually improve as new technology becomes available, and that way the trial doesn't lose relevance over time.

Matthew De Roos [07:20]

We're hopeful that these trials will provide results to support farmers in making better informed decisions, and options around what they can do to mitigate drier conditions.

Professor James Hunt [07:31]

The fortunes of rural communities really follows the fortunes of the farm businesses that are around them, and we have a predominance of broadacre farming in this district, so, if the broadacre farmers in this district are doing well, the communities tend to do well as well.

Robert Brown [07:50]

I think these types of long-term projects are really important. Profitability of farmers is directly linked to profitability of communities and the whole national economy.

Professor James Hunt [08:00]

From a global perspective, we've really come to a crunch point in terms of how we produce food for our planet. Between now and 2050, we've still got the most rapid period of population growth that there ever has been on the planet, and we need to produce enough food for those people. So, at a global level, that's really what we've got to achieve.

Now, individually at a farm business level, growers need to have sustainable farm businesses and they need to be able to regularly produce profits and get return on capital, and do that without using practices that damage the wider environment and natural resources, and the way they do that is by looking at research like this and the evidence that we and others can produce to adopt the things that they need to in order to meet all of those goals.

Rhiannan McPhee [08:48]

I think farmers have to make a lot of decisions, when you look at the seasons, when you look at when to or to not do something, these trials can take some of that risk away because there's data behind them, so they've got more confidence in, you know, what they're going to do on a day-to-day basis and prepare for tough times like drought.

Professor James Hunt [09:05]

We hope that the findings will influence drought resilience strategies by helping growers better understand which ratios and the sorts of enterprises they should be using in their region, that allows them to move their business towards a structure that is inherently more drought-resilient.

Recording ends [9:32]

A screen displays the logos for the Department of Agriculture, Fisheries and Forestry and the Future Drought Fund.

Acknowledgement of Country

We acknowledge the continuous connection of First Nations Traditional Owners and Custodians to the lands, seas and waters of Australia. We recognise their care for and cultivation of Country. We pay respect to Elders past and present, and recognise their knowledge and contribution to the productivity, innovation and sustainability of Australia's agriculture, fisheries and forestry industries.

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This project was funded under the [Future Drought Fund's Long-term Trials of Drought Resilient Farming Practices Grants program](#), and delivered by the University of Melbourne.

Source: <https://www.agriculture.gov.au/about/news/mixed-farming-strategies-drier-future>